Tuesday, January 18, 2022 10:38 AM

1. 32 r → 8 biți

2. LDi SUBi → R_{31}-R_{16}

3. LDI R_0-R_{15}
LDI R_{31}-R_{16}
MOV R_0, R_{15}

4. BREQ/BRNE → BRBS/BRBC
(salt instrucțiuni)

5. I/O memory → configurare și interogare periferice

6. I/O instructions → IN/out, SBI/CBI,

7. I - O address → 0x00 - 0x3F
SRAM → 0x40 - 0xBF

8. → ISA nu există
→ scriere PC JMP BRBS
→ citire RCALL + POP + POP

9. f_{new} ← f_{CLK} / D (prescaler)

10. RETI → RET + SEI /RCALL-ISR → RCALL+CLI

11. Vector de întreruperi → reți că nu este implementată procedura de tratare a întreruperii

12. http://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-8235-8-bit-AVR-Microcontroller-ATtiny20_Datasheet.pdf

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COM0B1 — 1
COM0B0 — 0

72 WGM0 2/1/0 → 111
[2:0]

73 CS0 [2:0] → 010

0x19 ← 69 TCCR0A 00 10 00 11 → 0x23

0x18 ← 72 TCCR0B 00 00 10 10 → 0x0A

0x16 ← T2TCCR0B 00 00 1010 → 0x0A

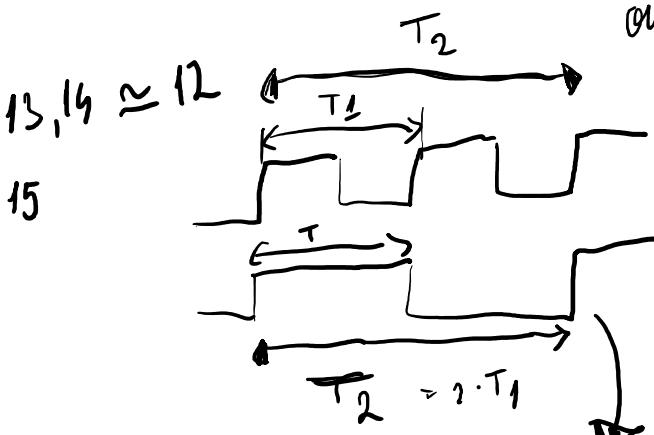
pg 74 { 0x16 OCR0A → 130
0x15 OCR0B → 34

tA equ 0x19

ldi R16, 0x23
out 0x19, R16
ldi R16, 0x0A
out 0x18, R16
ldi R16, 130
out 0x16, R16
ldi R16, 34
out 0x15, R16

Hulk  . l pg 73 TCNT0 0x17

ldi R16, 0
out 0x17, R16

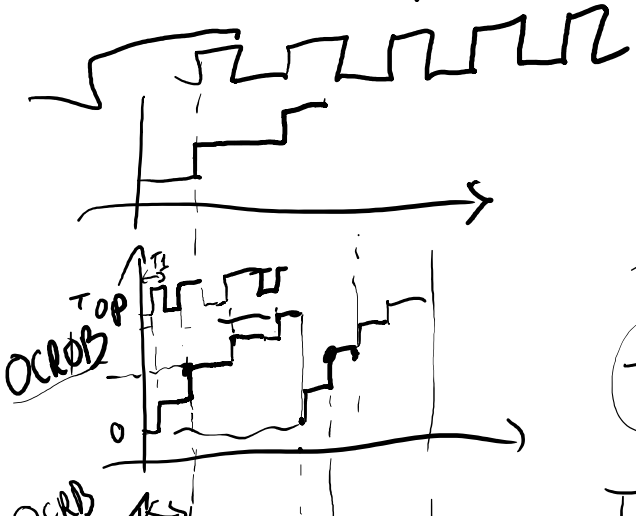


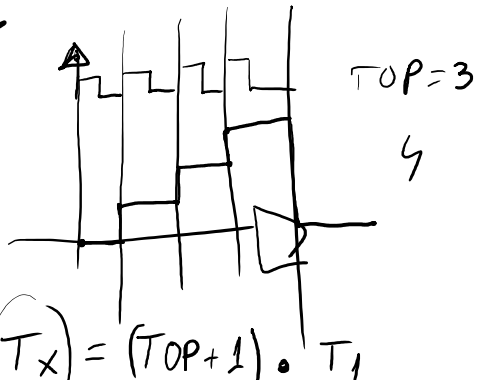
13, 14 ≃ 12

15

INPUT CLK

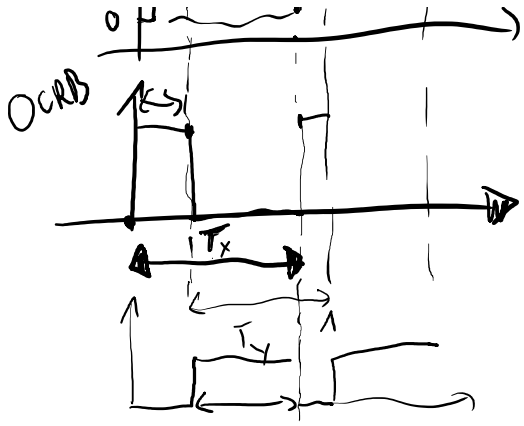
D = 2





$T_x = (TOP+1) \cdot T_1$

T = D



$$T_1 = \frac{D}{F_{CLK}}$$

$$F_1 = F_{CLK} / D$$

$$T_{CLK} = \frac{1}{F_{CLK}}$$

$$T_1 = T_{CLK} \cdot D$$

$$T_x = (TOP+1) \cdot \frac{D}{F_{CLK}}$$

$$T_x \cdot F_{CLK} = (TOP+1) \cdot D$$

$$OCR\phi B = \frac{TOP+1}{4}$$

$$0.04096 \cdot 10^{-3} \cdot 100 \cdot 10^{6} = 4096$$

$$(TOP+1) \cdot D = 4096$$

$$D = 2^{X}, \quad X \in 10$$

$D = 1024$	$TOP = 3$
$D = 256$	$TOP = 15$
$D = 64$	$TOP = 63$
$D = 8$	$TOP = 511$

TOP — 8 bit

$D = 1024$ $\quad OCR\phi B = 1$

$CS\phi[2:0] = 101 \simeq 12$